

A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-123482119})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K , the normalized generator σ_x , the pair (a', J) representing the torsion class e_j , the auxiliary divisor I' , the compactly represented element t , a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1 - \sigma_x)^2 I' + \text{div}(t) + i_x(J) = 0, \quad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$ -basis of the ideal. For example, the ideal $J = \begin{pmatrix} 2306 & 1001 \\ 0 & 1 \end{pmatrix}$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z} 2306 + \mathbf{Z} (1001 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Which integral basis. The one PARI returns for the field in question – and that basis is LLL-reduced, so it is not canonical: another machine may reduce differently, and the same coordinate vector would then denote a different algebraic number. The certificate therefore records the basis it was written against, as its last field per entry and in the header for K . The verifier expresses the stored basis in its own, checks that the resulting matrix is integral with determinant ± 1 – so that both bases span the same ring of integers – and converts the coordinates before checking anything. Where the two bases agree, that matrix is the identity and nothing changes.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\text{Cl}(K)$ (as HNF ideals). A character x on $\text{Cl}(K)/5$ is recorded by its value vector $(x(\bar{\kappa}_1), x(\bar{\kappa}_2), x(\bar{\kappa}_3))$; thus $(1, 0, 0)$ is the character dual to $\bar{\kappa}_1$. The column number j records which basis element e_j of $\text{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x . Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x .

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 - s + 30870530 = 0$, of discriminant -123482119 . Class group invariants $(270 \ 5 \ 5)$, class number 6750; the three stored generator ideals (HNF) are $\begin{pmatrix} 3701 & 407 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 5321 & 3325 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 5485 & 455 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a , column e_1

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + 3143y^3 + (13s + 203277)y^2 + (805s - 2808077)y + (3948s + 180371199)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 + 6290y^8 + 393995y^7 + 3449966y^6 + 1649817125y^5 \\ & + 28170766301y^4 + 638438154704y^3 + 104390489104063y^2 - 8166365 \\ & 85793047y + 33015651347624373 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ 0 \ 1 \ 0 \ -1 \ 0 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{101845979}{32603672280655888} + \left(-\frac{27721}{65207344561311776}\right)s, \quad J = \begin{pmatrix} 2306 & 1001 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2306$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $301379643833787 = 3^4 \cdot 3720736343627$:

$$\begin{pmatrix} 11162209030881 & 9694892882580 & 9429338425302 & 10539739703127 & 728631965825 & 4983667736004 & 484741371976 & 341684204281 & 7940809015466 & 7691241836001 \\ 0 & 3 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 3 & 0 & 0 & 1 & 1 & 2 & 1 & 2 \\ 0 & 0 & 0 & 3 & 2 & 2 & 2 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 4 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 2: character a , column e_2

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + 3143y^3 + (13s + 203277)y^2 + (805s - 2808077)y + (3948s + 180371199)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 + 6290y^8 + 393995y^7 + 3449966y^6 + 1649817125y^5 \\ & + 28170766301y^4 + 638438154704y^3 + 104390489104063y^2 - 8166365 \\ & 85793047y + 33015651347624373 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ 0 \ 1 \ 0 \ -1 \ 0 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{2039821869}{4265464134642730601} + \left(-\frac{58240}{4265464134642730601}\right)s, \quad J = \begin{pmatrix} 5321 & 3325 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5321$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $183134042912161 = 11 \cdot 12721 \cdot 1308745331$:

$$\begin{pmatrix} 183134042912161 & 20800793765058 & 138973395532151 & 1815266478758 & 117730674209745 & 130437536825493 & 124815487559744 & 50948764649726 & 114660011809634 & 149276735157783 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 2; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 0 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 3: character a , column e_3

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + 3143y^3 + (13s + 203277)y^2 + (805s - 2808077)y + (3948s + 180371199)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 + 6290y^8 + 393995y^7 + 3449966y^6 + 1649817125y^5 \\ & + 28170766301y^4 + 638438154704y^3 + 104390489104063y^2 - 8166365 \\ & 85793047y + 33015651347624373 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ 0 \ 1 \ 0 \ -1 \ 0 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{601351253}{4964587386703928125} + \left(-\frac{386132}{4964587386703928125}\right)s, \quad J = \begin{pmatrix} 5485 & 455 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5485$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $113043947459949 = 3^4 \cdot 11^3 \cdot 157 \cdot 331 \cdot 20177$:

$$\begin{pmatrix} 103805277741 & 79528556151 & 60665060555 & 14789234891 & 95688952133 & 19692499249 & 45724492741 & 75503706402 & 16594326331 & 57506767355 \\ 0 & 99 & 11 & 27 & 46 & 95 & 62 & 88 & 71 & 27 \\ 0 & 0 & 11 & 0 & 1 & 4 & 2 & 4 & 0 & 10 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 2; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (0 \ 4 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 4: character b , column e_1

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3y^4 + (s - 809)y^3 + (5s + 42600)y^2 + (-1098s - 3684779)y + (-10350s - 58460997)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 6y^9 - 1608y^8 + 90056y^7 + 23897931y^6 + 144996679y^5 - 58894974673y^4 - 1197446954596y^3 + 42622566601918y^2 + 1132577963924682y + 6725221591477959$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{101845979}{32603672280655888} + \left(-\frac{27721}{65207344561311776}\right)s, \quad J = \begin{pmatrix} 2306 & 1001 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2306$. **Checked:** $(a')^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $128889748995233 = 2557 \cdot 159499 \cdot 316031$:

$$\begin{pmatrix} 128889748995233 & 40302964010751 & 107693741560834 & 37161965696106 & 118244994313278 & 30446808581562 & 95073237225944 & 110267200967319 & 53139033007834 & 102926644032108 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 2; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 0 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 5: character b , column e_2

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3y^4 + (s - 809)y^3 + (5s + 42600)y^2 + (-1098s - 3684779)y + (-10350s - 58460997)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 6y^9 - 1608y^8 + 90056y^7 + 23897931y^6 + 144996679y^5 - 58894974673y^4 - 1197446954596y^3 + 42622566601918y^2 + 1132577963924682y + 6725221591477959$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{2039821869}{4265464134642730601} + \left(-\frac{58240}{4265464134642730601}\right) s, \quad J = \begin{pmatrix} 5321 & 3325 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5321$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $716420421540337 = 41 \cdot 3701 \cdot 4721337157$:

$$\begin{pmatrix} 716420421540337 & 439915674276760 & 486577769761015 & 378703171675192 & 54620729612907 & 426757902429438 & 568713151003147 & 3560740732264 & 559693418986395 & 62380647010835 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 0 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 6: character b , column e_3

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3y^4 + (s - 809)y^3 + (5s + 42600)y^2 + (-1098s - 3684779)y + (-10350s - 58460997)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 6y^9 - 1608y^8 + 90056y^7 + 23897931y^6 + 144996679y^5 - \\ & 58894974673y^4 - 1197446954596y^3 + 42622566601918y^2 + 11325779 \\ & 63924682y + 6725221591477959 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{601351253}{4964587386703928125} + \left(-\frac{386132}{4964587386703928125}\right) s, \quad J = \begin{pmatrix} 5485 & 455 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5485$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $773848489785849 = 3^2 \cdot 131321 \cdot 654755641$:

$$\begin{pmatrix} 257949496595283 & 249684560401265 & 92865342864585 & 88881048392873 & 94241407971941 & 206258727450266 & 229796109771003 & 95221440722517 & 81115071193515 & 61039465987735 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 2 & 2 & 2 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 7 factors with signed exponents of absolute value up to 2; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (0 \ 0 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 7: character c , column e_1

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 2038y^3 + (-14s + 7)y^2 + 80212y + (28286s - 14143)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 4076y^8 + 4313868y^6 + 6377567943y^4 - 24443272561294y^2 + 24699442814327431$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \ -2 \ 1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{101845979}{32603672280655888} + \left(-\frac{27721}{65207344561311776}\right)s, \quad J = \begin{pmatrix} 2306 & 1001 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2306$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1018586032545285 = 3^2 \cdot 5 \cdot 29^2 \cdot 43 \cdot 163 \cdot 3840017$:

$$\begin{pmatrix} 11707885431555 & 8897715785590 & 8731229078173 & 8967021363412 & 4428454790729 & 6458710182158 & 7347614000588 & 5456952104405 & 6953438502534 & 9992979416085 \\ 0 & 29 & 26 & 19 & 8 & 18 & 16 & 9 & 24 & 3 \\ 0 & 0 & 3 & 1 & 1 & 2 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 324 factors with signed exponents of absolute value up to 2852864096420; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 8: character c , column e_2

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 2038y^3 + (-14s + 7)y^2 + 80212y + (28286s - 14143)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 4076y^8 + 4313868y^6 + 6377567943y^4 - 24443272561294y^2 + 24699442814327431$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \ -2 \ 1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{2039821869}{4265464134642730601} + \left(-\frac{58240}{4265464134642730601}\right)s, \quad J = \begin{pmatrix} 5321 & 3325 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5321$. **Checked:** $(a')^J = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $53073656002475 = 5^2 \cdot 43 \cdot 71 \cdot 695363983$:

$$\begin{pmatrix} 53073656002475 & 26349265848645 & 52254118441753 & 5162972146874 & 13650257953896 & 33232368743957 & 40154191976762 & 39969065660670 & 22965542274587 & 5470058977411 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 328 factors with signed exponents of absolute value up to 2125204679829; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 9: character c , column e_3

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 2038y^3 + (-14s + 7)y^2 + 80212y + (28286s - 14143)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 4076y^8 + 4313868y^6 + 6377567943y^4 - 24443272561294y^2 + 24699442814327431$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \ -2 \ 1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{601351253}{4964587386703928125} + \left(-\frac{386132}{4964587386703928125}\right)s, \quad J = \begin{pmatrix} 5485 & 455 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5485$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $167516208537215 = 5 \cdot 187669 \cdot 178523047$:

$$\begin{pmatrix} 167516208537215 & 63543303819781 & 40891670051371 & 163132407498721 & 105857733906588 & 59150729847957 & 105987569427054 & 79278162390118 & 158050259564401 & 112876643350814 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 325 factors with signed exponents of absolute value up to 1528099554654; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 1 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + 2477y^3 + (s + 99677)y^2 + (-547s - 2176452)y + (25432s + 116411021)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 + 4955y^8 + 194401y^7 + 1582723y^6 + 731003260y^5 - \\ & 1049871065y^4 + 109049721499y^3 + 38754726953157y^2 - 1365743682 \\ & 252975y + 33521132255139233 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{101845979}{32603672280655888} + \left(-\frac{27721}{65207344561311776}\right)s, \quad J = \begin{pmatrix} 2306 & 1001 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2306$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $954790963349567 = 71^2 \cdot 189405071087$:

$$\begin{pmatrix} 13447760047177 & 7301202517667 & 5099431484620 & 1726471814904 & 9561197763960 & 7957130067557 & 7711065935835 & 638629510102 & 6546143098024 & 6232563125703 \\ 0 & 71 & 47 & 49 & 10 & 25 & 26 & 63 & 3 & 50 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 1080 factors with signed exponents of absolute value up to 25310787346386697130381438695021134989048594; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \ 4 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + 2477y^3 + (s + 99677)y^2 + (-547s - 2176452)y + (25432s + 116411021)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2y^9 + 4955y^8 + 194401y^7 + 1582723y^6 + 731003260y^5 - 1049871065y^4 + 109049721499y^3 + 38754726953157y^2 - 1365743682252975y + 33521132255139233$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{2039821869}{4265464134642730601} + \left(-\frac{58240}{4265464134642730601}\right)s, \quad J = \begin{pmatrix} 5321 & 3325 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5321$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $705111402369537 = 3^2 \cdot 71 \cdot 424093 \cdot 2601931$:

$$\begin{pmatrix} 235037134123179 & 54781051842312 & 118746788783711 & 67082409614250 & 48814698257518 & 204769882102508 & 88737827282103 & 21600865656612 & 229950878499927 & 48241865615620 \\ 0 & 3 & 0 & 1 & 2 & 2 & 0 & 0 & 2 & 2 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 1085 factors with signed exponents of absolute value up to 12167129602129086946573171804864191419967684; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 0 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + 2477y^3 + (s + 99677)y^2 + (-547s - 2176452)y + (25432s + 116411021)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2y^9 + 4955y^8 + 194401y^7 + 1582723y^6 + 731003260y^5 - 1049871065y^4 + 109049721499y^3 + 38754726953157y^2 - 1365743682252975y + 33521132255139233$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{601351253}{4964587386703928125} + \left(-\frac{386132}{4964587386703928125}\right)s, \quad J = \begin{pmatrix} 5485 & 455 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5485$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $417818487826197 = 3^2 \cdot 46424276425133$:

$$\begin{pmatrix} 139272829275399 & 88713202440596 & 92755567741182 & 90177769625912 & 130026083504969 & 8390862915365 & 34713384729554 & 17208681903498 & 64312609986576 & 50418878540397 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 2 & 2 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 1085 factors with signed exponents of absolute value up to 11687433564430013207989635317620972294978905; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \ 4 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 4223y^3 + (32s - 16)y^2 - 2600016y + (-2304s + 1152)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 8446y^8 + 12633697y^6 + 9651687328y^4 + 2208038365440y^2 + 163873614053376$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{101845979}{32603672280655888} + \left(-\frac{27721}{65207344561311776}\right)s, \quad J = \begin{pmatrix} 2306 & 1001 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2306$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $9464744484864 = 2^{13} \cdot 3^4 \cdot 257 \cdot 55501$:

$$\begin{pmatrix} 684660336 & 222154512 & 650284596 & 683220186 & 175996950 & 234377421 & 130277844 & 520380619 & 111932259 & 323728374 \\ 0 & 24 & 16 & 8 & 22 & 8 & 0 & 1 & 15 & 20 \\ 0 & 0 & 4 & 2 & 0 & 2 & 0 & 2 & 0 & 3 \\ 0 & 0 & 0 & 6 & 0 & 0 & 4 & 4 & 4 & 4 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 4 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 442 factors with signed exponents of absolute value up to 43099987836648487482; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \ 1 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 4223y^3 + (32s - 16)y^2 - 2600016y + (-2304s + 1152)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 8446y^8 + 12633697y^6 + 9651687328y^4 + 2208038365440y^2 + 163873614053376$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{2039821869}{4265464134642730601} + \left(-\frac{58240}{4265464134642730601}\right)s, \quad J = \begin{pmatrix} 5321 & 3325 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5321$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $25775647135776 = 2^5 \cdot 3^2 \cdot 4621 \cdot 19367837$:

$$\begin{pmatrix} 1073985297324 & 5225662018 & 630948454836 & 1062010302135 & 354200203553 & 416677103064 & 535373255772 & 421369491208 & 285924777230 & 895865301630 \\ 0 & 2 & 0 & 0 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 12 & 11 & 8 & 7 & 7 & 0 & 9 & 9 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 442 factors with signed exponents of absolute value up to 40051445026885797606; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \ 0 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 4223y^3 + (32s - 16)y^2 - 2600016y + (-2304s + 1152)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 8446y^8 + 12633697y^6 + 9651687328y^4 + 2208038365440y^2 + 163873614053376$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{601351253}{4964587386703928125} + \left(-\frac{386132}{4964587386703928125}\right)s, \quad J = \begin{pmatrix} 5485 & 455 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5485$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $91146205970432 = 2^{14} \cdot 5563122923$:

$$\begin{pmatrix} 356039867072 & 252209300936 & 66898099604 & 18048622616 & 229209357714 & 236761310544 & 2098126469 & 310097952939 & 319692912162 & 156875209885 \\ 0 & 8 & 4 & 0 & 3 & 4 & 4 & 1 & 0 & 0 \\ 0 & 0 & 4 & 0 & 3 & 3 & 0 & 3 & 3 & 0 \\ 0 & 0 & 0 & 8 & 0 & 0 & 6 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 437 factors with signed exponents of absolute value up to 30658864764070769427; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (4 \ 2 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 1594y^3 + (32s - 16)y^2 - 2525895y + (-2592s + 1296)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 3188y^8 - 2510954y^6 + 39663975724y^4 + 1259095111857y^2 + 207402542786304$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-2 \ 2 \ -1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{101845979}{32603672280655888} + \left(-\frac{27721}{65207344561311776}\right)s, \quad J = \begin{pmatrix} 2306 & 1001 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2306$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $80312990617728 = 2^7 \cdot 3^4 \cdot 11 \cdot 101 \cdot 6972311$:

$$\begin{pmatrix} 557729101512 & 349832153790 & 462355049748 & 274706643287 & 342513048316 & 365118894658 & 111611279999 & 355101108901 & 50239291891 & 23562060756 \\ 0 & 18 & 0 & 1 & 2 & 0 & 8 & 4 & 10 & 9 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 254 factors with signed exponents of absolute value up to 4061008623; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \ 3 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 1594y^3 + (32s - 16)y^2 - 2525895y + (-2592s + 1296)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 3188y^8 - 2510954y^6 + 39663975724y^4 + 1259095111857y^2 + 207402542786304$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-2 \ 2 \ -1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{2039821869}{4265464134642730601} + \left(-\frac{58240}{4265464134642730601}\right)s, \quad J = \begin{pmatrix} 5321 & 3325 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5321$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $461557383369984 = 2^8 \cdot 3^{10} \cdot 11^2 \cdot 252341$:

$$\begin{pmatrix} 1798686648 & 666047448 & 607042272 & 655620738 & 108376403 & 37304866 & 299819624 & 970450028 & 1288038357 & 920414049 \\ 0 & 1782 & 1716 & 501 & 709 & 859 & 488 & 369 & 599 & 294 \\ 0 & 0 & 12 & 6 & 8 & 9 & 2 & 9 & 4 & 6 \\ 0 & 0 & 0 & 6 & 3 & 1 & 4 & 3 & 5 & 2 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 259 factors with signed exponents of absolute value up to 2003087558; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (1 \ 1 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 1594y^3 + (32s - 16)y^2 - 2525895y + (-2592s + 1296)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 3188y^8 - 2510954y^6 + 39663975724y^4 + 1259095111857y^2 + 207402542786304$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-2 \ 2 \ -1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{601351253}{4964587386703928125} + \left(-\frac{386132}{4964587386703928125}\right)s, \quad J = \begin{pmatrix} 5485 & 455 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5485$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $457970075887104 = 2^9 \cdot 3^2 \cdot 11^3 \cdot 74670073$:

$$\begin{pmatrix} 19712899272 & 7067514300 & 19703872672 & 12268240548 & 5325940829 & 78057772 & 2095614532 & 17306151784 & 6427559407 & 5934028045 \\ 0 & 132 & 44 & 76 & 66 & 44 & 56 & 33 & 120 & 126 \\ 0 & 0 & 22 & 0 & 0 & 8 & 0 & 21 & 12 & 6 \\ 0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 258 factors with signed exponents of absolute value up to 3995280885; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \ 3 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.